

Where does ISAC (immune-stimulating antibody conjugates) go from here?

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ABSTRACT

Immune-stimulating antibody conjugates (ISACs) have emerged as a promising class of therapeutics, employing design principles similar to antibody-drug conjugates (ADCs) but replacing cytotoxic payloads with immune-stimulatory agents. These agents enhance immune system activation, enabling more effective tumor targeting and elimination. However, recent clinical setbacks since 2021 have raised concerns about ISACs' viability as ADC successors. Although ISAC development remains early-stage, clinical trials have revealed significant challenges, including limited efficacy, systemic toxicity (eg, cytokine release syndrome), and anti-drug antibody generation. Current strategies to overcome these hurdles focus on optimizing payload selection (eg, toll-like receptor 7 (TLR7) vs stimulator of interferon genes (STING) agonists), improving linker stability, and exploring localized delivery methods. While ISACs face substantial translational barriers, their potential to synergize with existing immunotherapies and treat "cold" tumors maintains therapeutic promise. Further refinement of design parameters may ultimately position ISACs as a transformative oncology modality.

ISACs represent an emerging class of targeted immunotherapeutics designed to overcome limitations of conventional oncology modalities. Structurally analogous to ADCs, ISACs comprise a tumor-targeting monoclonal antibody, a chemical linker, and a payload of immune-stimulatory agents (eg, TLR 7/8 or STING agonists) rather than cytotoxic drugs. Unlike ADCs, which directly kill antigen-expressing tumor cells on internalization, ISACs leverage their payloads to reprogram the tumor microenvironment (TME).¹ By activating innate immune cells (eg, dendritic cells, macrophages) and enhancing antigen presentation, ISACs convert immunologically "cold" tumors into "hot" ones, thereby initiating systemic antitumor immunity and potential long-term immunological memory. This mechanistic shift aims to address key challenges in solid tumor treatment, including tumor heterogeneity and immunosuppressive TMEs, positioning ISACs as a strategic approach for enhancing response rates to immunotherapies.

While structurally analogous (antibody-linker-payload), ADCs and ISACs represent divergent therapeutic strategies. ADCs function as targeted cytotoxic delivery systems, whereas ISACs act as targeted immune system activators. This mechanistic divergence underpins their contrasting clinical profiles: ADCs contend with payload-driven cytotoxicity, while ISACs grapple with immune activation-related toxicities and immunogenicity, presenting distinct challenges for their clinical optimization and therapeutic niches.

The [figure 1](#) provides a comprehensive framework for understanding ISACs' dual mechanisms and inherent challenges. Unlike ADCs, which rely solely on payload internalization and direct cytotoxicity, ISACs synergize three distinct processes: targeted immune recruitment: antibody-mediated Antibody-dependent cell-mediated cytotoxicity/Antibody-dependent cellular phagocytosis (ADCC/ADCP) rapidly engages innate immunity (eg, macrophages phagocytosing opsonized tumor cells). Microenvironment reprogramming: local TLR/STING agonist release converts "cold" tumors by activating dendritic cells and inducing inflammatory cytokines (eg, interleukin (IL)-12 from macrophages). Systemic immunity: cross-primed CD8⁺ T cells establish durable anti-tumor memory.

Although the design strategies for different ISACs vary, they generally face a common challenge: severe toxicity associated with systemic administration. When drugs are administered systemically, they may trigger non-specific immune reactions, which can lead to potentially fatal cytokine storms. Some drug molecules have had to halt their research in clinical trials as a result. NJH395 is an ISAC targeting Human epidermal growth factor receptor 2 (HER2) for the treatment of non-breast HER2+malignant tumors. NJH395 is composed of IgG1 MIW338 and an immune-stimulatory TLR7 agonist. MIW338 is a humanized monoclonal antibody that targets HER2, belonging to the IgG1/κ isotype subclass and sharing a



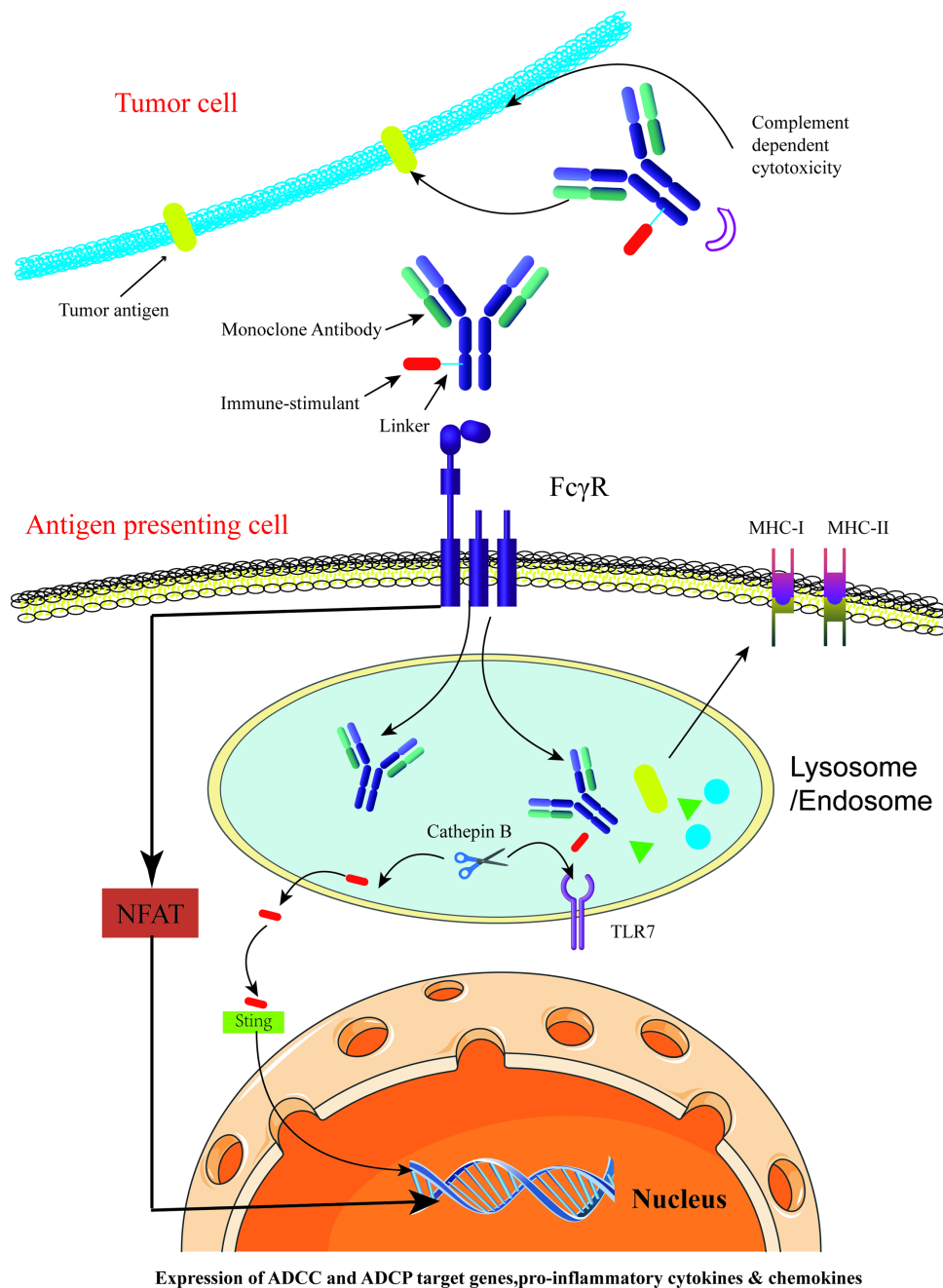
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Expression of ADCC and ADCP target genes, pro-inflammatory cytokines & chemokines

Figure 1 The ISACs potentiate antitumor immune responses. Tumor cell killing through antibody-mediated effector functions on specific recognition of tumor antigen by the mAb variable region and binding of the mAb Fc fragment by FcγR-expressing cells, such as antigen-presenting cells (activation of NFAT-mediated ADCC and ADCP), or by the complement ("complement-dependent cytotoxicity"), immune activation (eg, STING or TLR7) in the tumor target cells or neighboring myeloid cells after ISAC internalization. ISAC, immune-stimulating antibody conjugates; MHC, major histocompatibility complex; TLR, toll-like receptor; STING, stimulator of interferon gene; NFAT: Nuclear Factor of activated T Cell; ADCC, Antibody-dependent cell-mediated cytotoxicity; ADCP, Antibody-dependent cellular phagocytosis; mAb, Monoclonal Antibody

similar sequence with trastuzumab. Through genetic engineering techniques, four cysteine (Cys) residues were introduced into MIW338 at specific sites to avoid disrupting its binding with HER2 and FcγR. Site-specific conjugation was achieved by linking maleimide with a TLR7 agonist, using a non-cleavable linker to minimize the risk of payload release before binding to HER2+tumor cells, thereby providing stability, reducing

systemic exposure, and minimizing immune-related toxicity. NJH395 exhibits specific binding to human and cynomolgus monkey HER2 ($K_d=0.33/5.0$ nmol/L) and demonstrates a binding affinity similar to T-DM1, underscoring the clinical relevance of NJH395.²

When HER2+breast cancer cells HCC1954 were treated with NJH395 and MIW338, experimental results indicated that both compounds could bind to the target

cells, unlike their isotype control counterparts. Moreover, internalization of NJH395 within HCC1954 cells was observed, whereas internalization of the isotype control ISAC (NJH397) was not detected. Incubation of NJH395 with N87 gastric cancer cells *in vitro* resulted in the opening of the maleimide ring, suggesting that NJH395 undergoes lysosomal processing post-internalization to generate active metabolites, primarily comprising MIM697 (Cys-Linker-Payload) and a minor metabolite YFB712. Through *in vitro* assays measuring TLR activation effects using these metabolites, both metabolites induced the expression of the Nuclear factor kappa-B (NF- κ B) reporter gene in human embryonic kidney (HEK)-293T cells expressing TLR7, with a peak activation level equivalent to the TLR7/8 agonist resiquimod. Additionally, MIM697 did not exhibit significant activity in HEK-TLR8 reporter gene assays, suggesting specificity towards TLR7.

Due to TLR7 induction of pro-inflammatory cytokines (tumor necrosis factor (TNF)- α , IL-6, IL-8, and IL-1 β) transcription, the activation level of TLR7 can be evaluated by measuring these cytokines. Stimulation with MIM697 and YFB712 results in the secretion of interferon (IFN)- α and IL-6, demonstrating robust TLR7 activation in human peripheral blood mononuclear cells.³ Whole blood from healthy donors was co-incubated with NJH395 or MIW338. Quantitative Real-time PCR (qPCR) analysis revealed upregulation of TNF- α , IL-6, and IL-1 β genes in the NJH395-treated group, while no upregulation was observed in the MIW338-treated group. This suggests that this effect is driven by the stimulatory action of TLR7. Furthermore, the upregulation of OAS1 and OAS2 in the downstream signaling pathway of TLR7 provides additional support for this hypothesis. Co-culturing of HER2+HCC1954 tumor cells with HER2-THP-1 macrophage cells was used to further assess the therapeutic efficacy of NJH395. Following incubation with NJH395 or controls, the IL-1 β production in the supernatant was evaluated. NJH395 underwent a Phase I clinical trial in December 2018 (NCT03696771) for the treatment of non-breast HER2+malignant tumors. As the first clinical trial testing ISAC in humans, the results were disappointing. Data from 18 patients receiving a single-dose escalation showed that TLR7 agonists induce type I IFN response in HER2+tumor cells, which is related to immune modulation in the TME. However, at tolerated doses, efficacy was insufficient with an objective response rate of 0%. NJH395 induced rapid and robust systemic immune activation, with cytokine release syndrome being a common drug-related adverse event (55.6%, Grade $\geq$ 2). High-dose administration led to the development of ADA (anti-drug antibody) and neuroinflammation, posing significant clinical challenges.⁴ NJH395's non-cleavable linker failed to prevent Fc γ R-mediated toxicity, underscoring the need for tumor-restricted payload release.

SBT6050/6290 (Silverback), using a TLR8 agonist as the payload, also faced setbacks. Its discontinuation may be related to cytokine-related toxic side effects. SBT6050 underwent two clinical trials (NCT05091528/

NCT04460456) for the treatment of HER2+solid tumors. NCT05091528 (Phase I/II) was terminated due to considerations of portfolio prioritization. NCT04460456 (Phase I) aimed to evaluate the safety, tolerability, and activity of subcutaneous injections of escalating doses of SBT6050, either alone or in combination with programmed cell death protein-1 (PD-1) checkpoint inhibitors (pembrolizumab/cemiplimab), for the treatment of locally advanced (unresectable) and/or metastatic HER2+solid tumors in adult patients. Results indicated poor treatment efficacy of SBT6050 at tolerated doses, with preliminary data across all dose levels showing partial response (n=1), stable disease (n=3), and disease progression (n=10), with a 7% partial response rate.⁵ SBT6290 targets the tumor-associated antigen nectin-4. A Phase I/II clinical trial for SBT6290 (NCT05234606), either as monotherapy or in combination with pembrolizumab for treating nectin-4+advanced solid tumors, was initiated in March 2022 but terminated due to portfolio prioritization. Clinical development of SBT6290 was discontinued prematurely due to portfolio prioritization—aligns mechanistically with unmanageable TLR8-driven cytokine release syndrome (CRS), suggesting strategic misjudgment of target biology.^{6,7} TLR8's inherent hyperinflammatory nature demands ultra-stable linkers and staggered dosing with ICIs.

In addition to the ISAC constructed by TLR receptor agonists, the ISAC constructed by STING agonists also faces a challenging future. On March 13, 2023, Mersana announced the cessation of the Phase I clinical trial of its ISAC pipeline XMT-2056 (NCT05514717) by the US Food and Drug Administration (FDA). This decision was made due to a Grade 5 (fatal) severe adverse event observed in the first dose group of XMT-2056 during the dose escalation trial. Subsequently, on November 1, 2023, Mersana revealed that the US FDA had removed the clinical hold on XMT-2056, allowing for the resumption of the Phase I dose escalation study targeting HER2+solid tumors. XMT-2056 was developed by Mersana using their dolaflexin technology platform, with a precise drug-to-antibody ratio (DAR) value of 8. Mersana's revised dosing protocol (0.1 mg/kg starting dose) may improve XMT-2056's safety profile. Ongoing clinical monitoring of XMT-2056 will determine whether design modifications successfully mitigate neurotoxicity risks. STING ISACs require blood-brain barrier shielding and primate-specific safety screens.

Certainly, not all ISACs have a bleak future. Developed by Bolt Biotherapeutics, BDC-1001's orphan drug designation reflects its unique PEGylated design, which shows reduced CRS incidence.⁸ BDC-1001 underwent Phase I/II clinical trials in February 2020 (NCT04278144) for the treatment of advanced HER2+solid tumors.⁹ In August 2023, Phase II clinical trials commenced (NCT05954143) for the use of BDC-1001 alone or in combination with pertuzumab

in patients with HER2+metastatic breast cancer previously treated with ENHERTU. Preclinical evidence indicates PEGylation in BDC-1001 reduces plasma-free agonist concentration by 78%, correlating with diminished cytokine release in primate models.^{10 11}

Although ISAC clinical trials have faced consecutive failures, there is no need to be overly disappointed. Consider that ADC was first proposed by Paul Ehrlich as early as 1979,¹² but due to the high technical barriers required for its synthesis and the long-term challenges of off-target effects and the discovery of specific antigens, it underwent years of experimentation and continuous iterations until ADC experienced rapid development in recent years.¹³ Therefore, returning to the pharmaceutical industry's famous saying that “the development of new drugs is inherently risky”, one can find solace.

The primary dose-limiting toxicities for first-generation ISAC conjugates mainly involve CRS and associated toxicities. These drugs exhibit suboptimal antitumor effects in patients, have narrow therapeutic windows, and some even develop ADAs. Potent immune activators combined with unstable linkers may lead to the shedding of drug payloads in the bloodstream, activating host immune cells and triggering CRS. Elevated DAR $\geq$ 8 increase off-target payload release, as quantified in STING-ISAC pharmacokinetic studies (DAR 8 vs 4: systemic agonist exposure+320%). Therefore, it is proposed to use moderately active agonists and stable linkers to reduce the release of small molecules in the peripheral circulation, prevent cytokine storms, and maximize the therapeutic window to enhance efficacy. The introduction of the ISAC concept has garnered significant attention from pharmaceutical companies regarding immune-modulating ADCs, yet the ability of ISAC to address the side effects of TLR agonists remains debatable.¹⁴ In simpler terms, enhancing therapeutic efficacy while constraining systemic toxicity presents greater challenges than mitigating established adverse effects.

In preclinical studies, ISAC has shown promising results. However, in clinical trials, its efficacy is poor, and systemic toxicity exists. On one hand, while increasing drug efficacy, toxicity will also increase, making it challenging to enhance efficacy while ensuring safety. On the other hand, the body can develop ADAs, accelerate metabolism, and affect the drug's efficacy. Several strategies may significantly help improve ISAC efficacy: (1) using purine-based TLR7 agonists to enhance ISAC's antitumor effects. In a preclinical study, ISAC, containing a purine-based TLR7 agonist linked to trastuzumab, was administered to healthy mice and mice with gastric cancer. This ISAC showed efficacy in mice with gastric cancer without causing weight loss in healthy mice.¹⁵ (2) The choice of TLR agonists affects ISAC efficacy. CL264-ISAC is more effective in activating myeloid antigen-presenting cells than T785-ISAC. T785 acts as a TLR7/8 agonist, whereas CL264 is specific

to TLR7.¹⁶ Exploring additional immune stimulants could lead to the development of new, effective ISACs. (3) ISAC can lead to the production of ADAs, affecting drug efficacy. The production of ADAs depends on the immunogenicity of ISAC antibodies. ADAs develop in 15–30% of patients receiving first-gen ISACs, reducing serum half-life by 40–60%. Alternative immune stimulants—including RIG-I agonists with restricted activation profiles—show preclinical potential to minimize immunogenicity. (4) Due to the specificity of immune stimulation, intratumoral administration can reduce systemic leakage of stimulants, thereby minimizing off-target toxicity. Advances in site-specific conjugation (eg, cysteine-engineered antibodies) enable precise DAR control. This strategy also requires lower doses to halt tumor progression. However, intratumoral injection faces technical challenges, including size, density, and accessibility. Furthermore, it is currently unclear whether administering drugs at a lesion site can provide systemic protection.

ISACs occupy a distinct niche in cancer immunotherapy by synergizing targeted antibody delivery with innate immune activation. Unlike ADCs—which rely on cytotoxic payloads for direct tumor killing—ISACs overcome the “cold tumor” limitation through immune-stimulatory payloads that remodel the TME. However, this mechanism introduces CRS risks, narrowing their therapeutic index compared with ADCs' manageable hematologic toxicities. Against bispecific antibodies (BsAbs), ISACs offer broader immune activation by engaging both antigen-presenting cells and T cells, whereas BsAbs primarily accelerate T-cell cytotoxicity. While BsAbs achieve rapid debulking in hematologic malignancies, ISACs require weeks for adaptive immune priming, limiting utility in aggressive cancers but showing superior potential for solid tumors via myeloid cell engagement. Relative to chimeric antigen receptor T (CAR-T) cells, ISACs provide critical off-the-shelf availability without complex manufacturing, yet lack CAR-T's durable memory responses. Strategically, ISACs may prime immunologically cold tumors for subsequent CAR-T therapy, mitigating T-cell exhaustion in solid tumors. When compared with cytokine therapies, ISACs localize cytokine release, thereby avoiding systemic vascular toxicity inherent to recombinant cytokines like IL-2. Nevertheless, ISACs' limited payload capacity constrains their immunostimulatory scope compared with cytokine therapies. The toxicity profiles further differentiate these modalities: ISACs' dose-limiting CRS mirrors BsAbs but diverges fundamentally from CAR-T neurotoxicity or ADC hematotoxicity. Consequently, ISACs excel in TME reprogramming for immunotherapy-resistant solid tumors but require modality-specific safety optimization. Their future integration with checkpoint inhibitors or ADC platforms may unlock synergistic potential unattainable with monotherapies.

In summary, ISACs represent a high-risk, high-reward therapeutic paradigm that uniquely bridges targeted

delivery and immune activation. While clinical setbacks (eg, NJH395's CRS, XMT-2056's neurotoxicity) underscore critical barriers—systemic toxicity, immunogenicity, and suboptimal efficacy—these challenges stem from definable mechanisms: payload leakage, agonist hyperactivation, and ADA-driven clearance. Three strategic imperatives emerge:

1. Engineering solutions: optimize linker-payload stability (eg, BDC-1001's PEGylation) and tumor-restricted agonists (purine-based TLR7).
2. Administration innovation: pursue intratumoral delivery to maximize local immune effects.
3. Combination synergy: sequence ISACs with checkpoint inhibitors or ADCs to leverage their “cold-to-hot” tumor conversion capability.

Future success demands primate-relevant safety screening, neoadjuvant trial validation, and target antigen stratification (eg, nectin-4>HER2 for myeloid engagement). With iterative refinement mirroring the ADC evolution trajectory, ISACs hold transformative potential for immunotherapy-resistant solid tumors, contingent on resolving their narrow therapeutic index.

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